MAT211: Calculus-II

Lecture Notes

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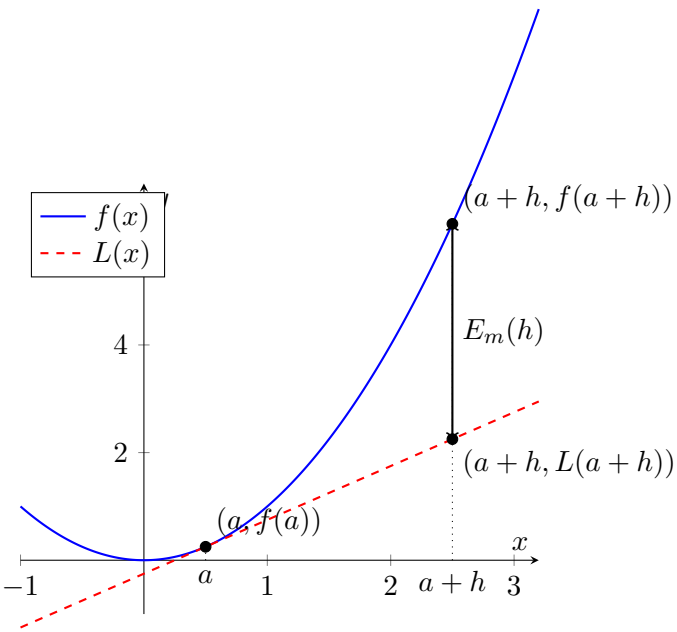
1 Lecture-01

Question: what did we mean by the derivative $f'(a)$?
Answer: Let U be an open subset of $\mathbb{R}$ and $f : U \to \mathbb{R}$ a function. Then f is differentiable at a, with derivative m, if and only if

$$\lim_{h\to 0} \frac{1}{h}\left(\underbrace{f(a+h)-f(a)}_{\Delta f} - \underbrace{mh}_{(*)}\right) = 0 \tag{1}$$

$(*)$: linear function of $\Delta x = h$.
Question: Why impose the domain U that needs to be an open set?
Answer: Write the answer.
The function mh that multiplies h by the derivative m is thus a linear function of the change in x. This definition emphasizes the idea that a function f is differentiable at a point a if the increment Δf to the function is well approximated by a linear function of the increment h to the variable, this linear function is $f'(a)h$. Let's see how we can confirm that:
Suppose $f(x)$ is our desired function and we want to approximate the function in the neighborhood at $x = a$. For this purpose, we will draw several straight lines through the point $(a, f(a))$ and see which line approximates the function best. Here, the best means, the linear line should approximate the function well at $x = a + h$. Suppose the slope of that linear function is m.



Then,

$$\frac{L(a+h)-L(a)}{a+h-a} = m$$
$$\frac{L(a+h)-f(a)}{h} = m$$
$$L(a+h) = f(a) + mh$$

Now, we want to minimize the error, $E_m(h) = f(a+h) - L(a+h)$. Hence, we need to find the optimal value for m, for which:

$$\frac{E_m(h)}{h} \to 0$$

$$\boxed{\text{optimal } m \iff \lim_{h\to 0} \frac{E_m(h)}{h} = 0}$$

A simple computation show that,

$$\lim_{h\to 0} \frac{E_m(h)}{h} = \lim_{h\to 0} \frac{f(a+h) - f(a) - mh}{h} = 0$$

which is the desired condition 1, we showed in the definition.

2 Lecture-02

Definition 2.1 (Euclidean Vector Space)**.** A Euclidean vector space is a finite-dimensional inner product space over $\mathbb{R}$.

Definition 2.2 (Inner product space)**.** An inner product space is a vector space V over the field F together with an inner product, that is, a map

$$\langle \cdot, \cdot \rangle : V \times V \to F$$

that satisfies the following three properties for all vectors $x, y, z \in V$ and all scalars $a, b \in F$.

1. (Symmetry) $\langle x, y \rangle = \langle y, x \rangle$.
2. (Linearity in the first argument) $\langle ax + by, z \rangle = a\langle x, z \rangle + b\langle y, z \rangle$.
3. (Positive-definiteness) If $x \neq 0$ then, $\langle x, x \rangle > 0$.

Example 2.3. Verify that the dot product function $\langle \cdot, \cdot \rangle : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ is an inner product.

To verify this we must prove the three conditions shown above. Let $u = (a, b)$ and $v = (c, d)$. Then,

1. $\langle u, v \rangle = ac + bd = ca + db = \langle v, u \rangle$.
2. Let $a, b \in \mathbb{R}$ and $u, v, w \in \mathbb{R}^2$ with $u = (u_1, u_2), v = (v_1, v_2), w = (w_1, w_2)$. Then

$$\begin{aligned}
\langle au + bv, w \rangle &= (au_1 + bv_1)w_1 + (au_2 + bv_2)w_2 \\
&= au_1w_1 + bv_1w_1 + au_2w_2 + bv_2w_2 \\
&= a(u_1w_1 + u_2w_2) + b(v_1w_1 + v_2w_2) \\
&= a\langle u, w \rangle + b\langle v, w \rangle.
\end{aligned}$$

3. For any $u = (u_1, u_2)$,

$$\langle u, u \rangle = {u_1}^2 + {u_2}^2.$$

   Since squares of non-zero numbers are positive, we have ${u_1}^2 > 0, {u_2}^2 > 0$ and therefore, $\langle u, u \rangle > 0$.

Definition 2.4 (Informal)**.** Define **norm** of a vector $x \in V$ as

$$||x|| = \sqrt{\langle x, x \rangle}.$$

Definition 2.5 (Metric Space)**.** A metric space is an ordered pair (M, d) where M is a set and d is a **metric** (distance) on M, i.e., a function

$$d : M \times M \to \mathbb{R}$$

satisfying the following axioms for all points $x, y, z \in M$:

1. $d(x, y) \geq 0$;
2. $d(x, y) = 0 \iff x = y$;

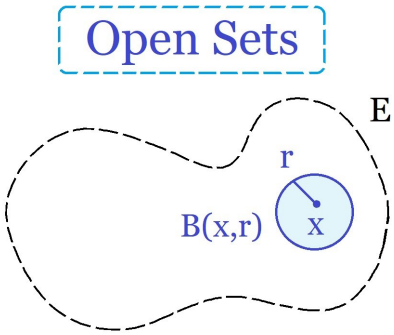
3. $d(x, y) = d(y, x)$.

4. $d(x, z) \leq d(x, y) + d(y, z)$.

Example 2.6 (Tax-cab Metric)**.** In $\mathbb{R}^2$, the taxi-cab distance between $p = (p_1, p_2)$ and $q = (q_1, q_2)$ is $|q_1 - p_1| + |q_2 - p_2|$. It is easy to verify that this is a metric.

Definition 2.7 (Open ball)**.** For any $x \in \mathbb{R}^n$ and any $r > 0$, the **open ball** of radius r around x is the subset

$$B_r(x) = \{y \in \mathbb{R}^n \mid |x - y| < r\}.$$



Definition 2.8 (Open sets of $\mathbb{R}^n$)**.** A subset $U \subseteq \mathbb{R}^n$ is open if for every point $x \in U$, there exists $r > 0$ such that the open ball $B_r(x)$ is contained in U.

Example 2.9. We prove that the Cartesian product $(a_1, b_1) \times (a_2, b_2)$ of two open intervals in $\mathbb{R}$ is an open set in $\mathbb{R}^2$ with the standard Euclidean topology.

Proof. Let d denote the Euclidean metric on $\mathbb{R}^2$:

$$d\big((x_1, y_1), (x_2, y_2)\big) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}.$$

Take an arbitrary point $p = (x, y) \in (a_1, b_1) \times (a_2, b_2)$. Then

$$a_1 < x < b_1, \qquad a_2 < y < b_2.$$

Define the four positive distances to the endpoints:

$$d_1 = x - a_1, \quad d_2 = b_1 - x, \quad d_3 = y - a_2, \quad d_4 = b_2 - y.$$

Let

$$r = \min\{d_1, d_2, d_3, d_4\} > 0.$$

We claim that the open ball $B(p, r) = \{q \in \mathbb{R}^2 : d(p, q) < r\}$ is contained in $(a_1, b_1) \times (a_2, b_2)$.

Indeed, take any point $q = (u, v) \in B(p, r)$. Then

$$|u - x| \leq d(p, q) < r, \qquad |v - y| \leq d(p, q) < r.$$

Hence

$$x - r < u < x + r, \qquad y - r < v < y + r.$$

Because $r \leq x - a_1$ and $r \leq b_1 - x$, we have

$$a_1 \leq x - r < u < x + r \leq b_1,$$

so $u \in (a_1, b_1)$. Similarly, $r \leq y - a_2$ and $r \leq b_2 - y$ imply

$$a_2 \leq y - r < v < y + r \leq b_2,$$

so $v \in (a_2, b_2)$. Consequently $q = (u, v) \in (a_1, b_1) \times (a_2, b_2)$.

Thus every point of the product has an open neighbourhood (an open ball) contained in the product. By definition, $(a_1, b_1) \times (a_2, b_2)$ is open in $\mathbb{R}^2$. $\square$

Definition 2.10 (Closed set)**.** A closed set A is defined by its ability to contain all of its limit points, meaning if a sequence of points $\{x_n\}$ within A converges to a limit x, then that limit x must also be in A.

Definition 2.11 (Sequence)**.** A sequence is a function whose domain is $\mathbb{N}$.

3 Lecture-03

Definition 3.1 (Sequence)**.** A sequence is a function where domain is $\mathbb{N}$.

Example 3.2. For example, let $x_n = \frac{1}{n}$ or $\left\{\frac{1}{n}\right\}_{n=1}^{\infty}$, be a sequence. Here, we write the sequence $f : \mathbb{N} \to \mathbb{R}$, where $f(n) = x_n = \frac{1}{n}$.

Definition 3.3 (Convergence of a Sequence)**.** A sequence of real numbers converges to a real number a if for every number $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that for all $n \geq N$

$$|a_n - a| < \varepsilon$$

Example 3.4. Take our earlier example 3.2 , prove that $x_n = \left\{\frac{1}{n}\right\}_{n=1}^{\infty}$ converges to 0.

Proof. Let, $\varepsilon > 0$, we choose $N \in \mathbb{N}$ such that $N > 1/\varepsilon$ (which is possible due to the Archimedean property), giving us $\frac{1}{N} < \varepsilon$.

Then, for all $n \geq N$ we have

$$\left|\frac{1}{n} - 0\right| \leq \left|\frac{1}{N} - 0\right| < \varepsilon$$

□

Example 3.5. Consider $x_{n+1} = \frac{1}{2}\left(x_n + \frac{2}{x_n}\right)$, $x_0 > 0$

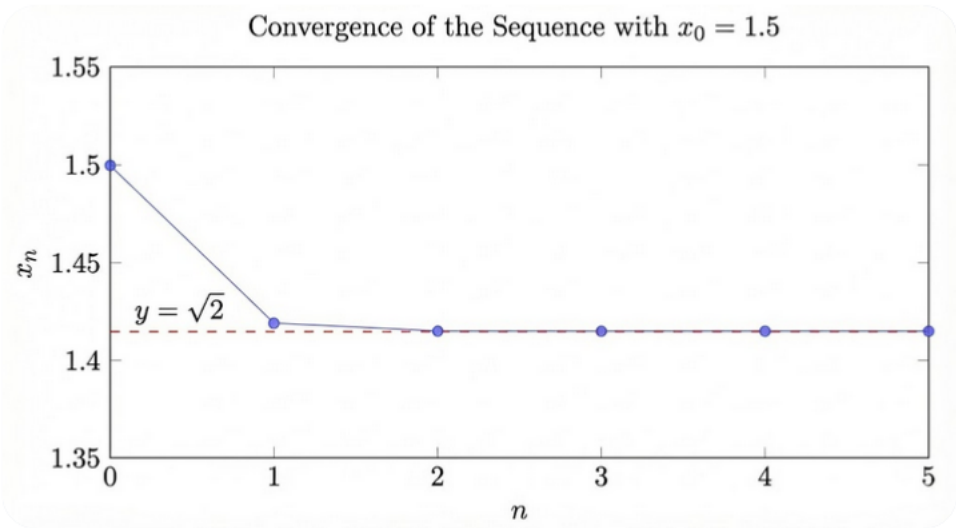
Show that x_n converges.

Proof. First, note that $x_n >= 0$ for all $n \in \mathbb{N}$. Because, $x_0 > 0$ and for induction, if we assume that $x_n > 0$ then,

$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{2}{x_n}\right) > 0$$

since all the terms in left side is positive.

Then note that, if you start abouve $\sqrt{2}$ you stay above and decrease. Refer to the picture below for an example.



Next, to rigorously proof that the limit exists. We can do so by proving this sequence to be monotone and bounded. We can prove that by testing both cases $0 < x_0 \leq \sqrt{2}$ and $0 < x_0 < \sqrt{2}$. Now, let $(x_n) \to L$, where $L > 0$. Then $(x_{n+1}) \to L$

$$\begin{aligned} L &= \lim_{n\to\infty} x_{n+1} \\ &= \lim_{n\to\infty} \frac{1}{2}\left(x_n + \frac{2}{x_n}\right) \\ &= \lim_{n\to\infty} \frac{x_n}{2} + \frac{1}{x_n} \\ &= \frac{L}{2} + \frac{1}{L} \end{aligned}$$

This gives us,

$$L = \frac{L}{2} + \frac{1}{L} \implies \frac{L}{2} = \frac{1}{L} \implies L = \sqrt{2}$$

Hence, the given sequence converges to $\sqrt{2}$. □

Definition 3.6 (Convergent sequence; limit of sequence for $\mathbb{R}^n$ Space)**.** A sequence $i \to a_i$ of points $\mathbb{R}^n$ converges to $a \in \mathbb{R}^n$ if

$$\forall \varepsilon > 0, \exists M | m > M \implies |a_m - a| < \varepsilon$$

We then call a, the *limit* of the sequence

This also holds for vector a_i.

$$a_m = [(a_m)_1, (a_m)_2, \cdots, (a_m)_n] \to [a_1, a_2, \cdots, a_n] \in \mathbb{R}^n$$

Here, a_1 is the convergent point of sequence $(a_m)_1, (a_{m+1})_1, \cdots$. And so on.

Proposition 3.7 (Convergence in terms of coordinates)**.** *A sequence $m \to a_m$ with $a_m \in \mathbb{R}^n$ convergers to a if and only if each coordinate;* i.e. *for all $1 \leq j \leq n$, the jth coordinate $(a_i)_j)$ of a_i converges to a_j, the jth coordinate of the limit a.*

Proof. Let $\|\cdot\|$ denote the standard Euclidean norm in $\mathbb{R}^n$. Recall the definition of sequence convergence: $a_m \to a$ means that for every $\epsilon > 0$, there exists an integer N such that for all $m \geq N$, $\|a_m - a\| < \epsilon$.

Forward Direction (⇒):
Assume $a_m \to a$. We want to show that for any fixed coordinate $j \in \{1, \ldots, n\}$, the sequence $(a_m)_j \to a_j$.

Let $\epsilon > 0$ be given. Because $a_m \to a$, there exists an integer N such that for all $m \geq N$:

$$\|a_m - a\| < \epsilon$$

Observe that the absolute value of any single coordinate difference is bounded by the Euclidean distance:

$$|(a_m)_j - a_j| = \sqrt{((a_m)_j - a_j)^2} \leq \sqrt{\sum_{k=1}^{n} ((a_m)_k - a_k)^2} = \|a_m - a\|$$

Therefore, for all $m \geq N$, we have:

$$|(a_m)_j - a_j| \leq \|a_m - a\| < \epsilon$$

Since for any $\epsilon > 0$ we found an N that satisfies this condition, $(a_m)_j \to a_j$ for each j.

Reverse Direction ($\Leftarrow$):
Assume that for each $1 \leq j \leq n$, the coordinate sequence $(a_m)_j \to a_j$. We want to show that the vector sequence $a_m \to a$.

Let $\epsilon > 0$ be given. By the definition of convergence for each coordinate, there exists an integer N_j such that for all $m \geq N_j$:

$$|(a_m)_j - a_j| < \frac{\epsilon}{\sqrt{n}}$$

We want this to hold for *all* coordinates at the same time, so we choose the largest threshold: let $N = \max\{N_1, N_2, \ldots, N_n\}$.

For any $m \geq N$, m is greater than or equal to every N_j, so the inequality $|(a_m)_j - a_j| < \frac{\epsilon}{\sqrt{n}}$ holds for all $j = 1, \ldots, n$ simultaneously.

Now, we evaluate the distance between the vectors a_m and a for $m \geq N$:

$$\|a_m - a\| = \sqrt{\sum_{j=1}^{n} ((a_m)_j - a_j)^2} < \sqrt{\sum_{j=1}^{n} \left(\frac{\epsilon}{\sqrt{n}}\right)^2}$$

Simplifying the expression under the square root:

$$\|a_m - a\| < \sqrt{\sum_{j=1}^{n} \frac{\epsilon^2}{n}} = \sqrt{n \cdot \frac{\epsilon^2}{n}} = \sqrt{\epsilon^2} = \epsilon$$

Because for any $\epsilon > 0$ we found an N such that $m \geq N \implies \|a_m - a\| < \epsilon$, the sequence a_m converges to a. $\square$

4 Lecture-08

4.1 Directional Derivative

Problem 4.1. If $T(\vec{x})$ is a linear transformation, then find the directional derivative of $T(\vec{x})$ at $\vec{x} = \vec{a}$ in the direction $\vec{v}$.

Problem 4.2. If $f : \mathbb{R}^n \to \mathbb{R}$ is defined by $f(\vec{x}) = \|\vec{x}\|^2$, find the directional derivative of f at $\vec{x} = \vec{a}$ in the direction $\vec{v}$.

Solution:

$$\begin{aligned}
D_{\vec{v}} f(\vec{a}) &= \lim_{t\to 0} \frac{f(\vec{a}+t\vec{v}) - f(\vec{a})}{t} \\
&= \lim_{t\to 0} \frac{\|\vec{a}+t\vec{v}\|^2 - \|\vec{a}\|^2}{t} \\
&= \lim_{t\to 0} \frac{(\vec{a}+t\vec{v})\cdot(\vec{a}+t\vec{v}) - \vec{a}\cdot\vec{a}}{t} \\
&= \lim_{t\to 0} \frac{2t\vec{a}\cdot\vec{v} + t^2\|\vec{v}\|^2}{t} \\
&= 2\vec{a}\cdot\vec{v}.
\end{aligned}$$

Example 4.3. Let

$$f(x,y) = \begin{cases} \dfrac{x^3}{x^2+y^2}, & (x,y) \neq (0,0), \\ 0, & (x,y) = (0,0), \end{cases}$$

Find the partial derivatives and directional derivatives of f at $(0,0)$.

Partial derivatives at $(0,0)$:

$$f_x(0,0) = \lim_{h\to 0} \frac{f(h,0) - f(0,0)}{h} = \lim_{h\to 0} \frac{\frac{h^3}{h^2} - 0}{h} = \lim_{h\to 0} \frac{h}{h} = 1.$$

$$f_y(0,0) = \lim_{k\to 0} \frac{f(0,k) - f(0,0)}{k} = \lim_{k\to 0} \frac{0-0}{k} = 0.$$

Directional derivative at $(0,0)$: Let $\vec{v} = (a,b) \neq (0,0)$. Then

$$D_{\vec{v}} f(0,0) = \lim_{t\to 0} \frac{f(ta,tb) - f(0,0)}{t} = \lim_{t\to 0} \frac{\frac{(ta)^3}{(ta)^2+(tb)^2}}{t} = \lim_{t\to 0} \frac{\frac{t^3a^3}{t^2(a^2+b^2)}}{t} = \frac{a^3}{a^2+b^2}.$$

Equivalently, if $\vec{u} = (u_1, u_2)$ is a unit direction, then

$$D_{\vec{u}} f(0,0) = \frac{u_1^3}{u_1^2+u_2^2} = u_1^3 \neq 0.$$

Question: Does the existence of all directional derivatives at a point imply the existence of the total derivative at that point? The answer is no, as shown by the above example. The function f has all directional derivatives at $(0,0)$, but it is not differentiable at $(0,0)$ because the limit in the definition of differentiability does not exist.

4.2 Tangent Space for $\mathbb{R}^2$

For a *fixed* direction v, the directional derivative at p defines a map

$$D_v\big|_p : C^\infty(\mathbb{R}^2) \longrightarrow \mathbb{R}, \qquad f \longmapsto D_v f(p),$$

which is linear. Moreover, it satisfies the Leibniz rule

$$D_v(fg)(p) = f(p)\, D_v g(p) + g(p)\, D_v f(p),$$

so $D_v|_p$ is a *derivation at* p (the "tangent vector as operator" viewpoint). The tangent space is *not* built from one fixed operator $D_v|_p$; rather it is built from the collection of *all* such derivations at p. A clean definition is

$$T_p\mathbb{R}^2 := \Big\{ X : C^\infty(\mathbb{R}^2) \to \mathbb{R} \;\Big|\; X \text{ is linear and } X(fg) = f(p)X(g) + g(p)X(f) \Big\}.$$

Elements of $T_p\mathbb{R}^2$ are operators that differentiate functions at p. Each direction $v = (v_1, v_2) \in \mathbb{R}^2$ produces an element

$$v \longmapsto D_v\big|_p \in T_p\mathbb{R}^2.$$

Define the standard derivations

$$\frac{\partial}{\partial x}\bigg|_p (f) = \frac{\partial f}{\partial x}(p), \qquad \frac{\partial}{\partial y}\bigg|_p (f) = \frac{\partial f}{\partial y}(p).$$

Then for $v = (v_1, v_2)$,

$$D_v\big|_p = v_1 \frac{\partial}{\partial x}\bigg|_p + v_2 \frac{\partial}{\partial y}\bigg|_p,$$

so $\left\{ \frac{\partial}{\partial x}\big|_p, \frac{\partial}{\partial y}\big|_p \right\}$ is a basis of $T_p\mathbb{R}^2$. A tangent vector at p can be viewed as an operator that takes a smooth function f and returns its directional rate of change at p. The tangent space $T_p\mathbb{R}^2$ is the vector space of all such derivative-operators (derivations), and in coordinates it has basis $\frac{\partial}{\partial x}\big|_p, \frac{\partial}{\partial y}\big|_p$.

Example 4.4. Let $f(x, y) = 3x - 2y$ and $p = (1, 1)$. For $v = (1, 0)$ and $w = (0, 1)$,

$$D_v\big|_p(f) = \frac{\partial f}{\partial x}(p) = 3, \qquad D_w\big|_p(f) = \frac{\partial f}{\partial y}(p) = -2.$$

For $u = (2, 5)$,

$$D_u\big|_p(f) = 2 \cdot 3 + 5 \cdot (-2) = 6 - 10 = -4.$$

Equivalently,

$$D_u\big|_p = 2 \frac{\partial}{\partial x}\bigg|_p + 5 \frac{\partial}{\partial y}\bigg|_p.$$

Example 4.5. Let $f(x, y) = x^2 + y^2$ and $p = (1, 2)$. Then

$$f_x = 2x, \qquad f_y = 2y \quad \Rightarrow \quad f_x(p) = 2, \quad f_y(p) = 4.$$

For $v = (3, -1)$,

$$D_v\big|_p(f) = 3 \cdot 2 + (-1) \cdot 4 = 6 - 4 = 2, \qquad \text{hence} \qquad D_{(3,-1)}\big|_p = 3\left.\frac{\partial}{\partial x}\right|_p - \left.\frac{\partial}{\partial y}\right|_p.$$

Example 4.6. From the previous example, for $v = (1, 0)$ and $w = (0, 1)$,

$$D_v\big|_p(f) = 2, \qquad D_w\big|_p(f) = 4.$$

Since $v + w = (1, 1)$,

$$D_{v+w}\big|_p(f) = 1 \cdot 2 + 1 \cdot 4 = 6, \qquad \text{and indeed} \qquad D_{v+w}\big|_p(f) = D_v\big|_p(f) + D_w\big|_p(f).$$

Example 4.7. Let $f(x, y) = x$, $g(x, y) = y^2$, $p = (1, 2)$, and $v = (3, -1)$. Then

$$D_v\big|_p(f) = 3, \qquad D_v\big|_p(g) = (-1) \cdot (2y)\big|_{(1,2)} = -4.$$

For the product $fg = xy^2$, compute

$$(xy^2)_x = y^2, \qquad (xy^2)_y = 2xy,$$

so at $(1, 2)$ we have $D(fg)(1, 2) = (4, 4)$ and hence

$$D_v\big|_p(fg) = (4, 4) \cdot (3, -1) = 12 - 4 = 8.$$

On the other hand,

$$f(p)\, D_v\big|_p(g) + g(p)\, D_v\big|_p(f) = (1)(-4) + (4)(3) = -4 + 12 = 8,$$

4.3 Total Derivative

Definition 4.8. A function $f : \mathbb{R} \to \mathbb{R}$ is called *differentiable at $x = a$* if there exists a real number L such that

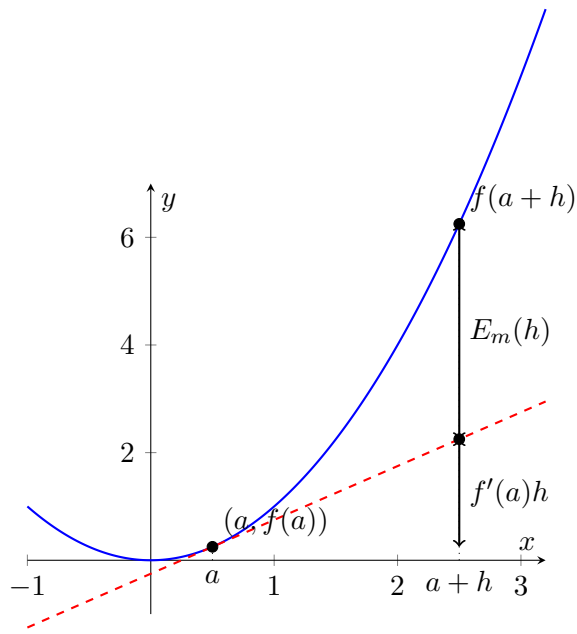
$$\lim_{h \to 0} \frac{f(a + h) - f(a) - Lh}{h} = 0.$$

Then the number L is called the *derivative* of f at $x = a$.

The definition indicates that,

$$f(a + h) = f(a) + Lh + o(h)$$

So, near $x = a$, the function f behaves like a linear function with slope L, $f(a+h) \approx f(a)+Lh =$ base point + linear correction. The derivative only captures the linear correction part. The constant term just shifts the graph vertically.



Remark: The notation $o(h)$ means that the error term $E(h) = f(a+h) - f(a) - Lh$ is small compared to h as $h \to 0$. More precisely, it means that $\lim_{h\to 0} E(h)/h = 0$. In other words, $E(h)$ goes to zero faster than h does. Formally, we say a function $E(h)$ is $o(h)$ if

$$\lim_{h\to 0} E(h)/h = 0.$$

In one variable $h^2 = o(h)$, $h^3 = o(h)$, but $2h$ is not $o(h)$ because $\lim_{h\to 0} 2h/h = 2 \neq 0$.

Definition 4.9. A function $f : \mathbb{R}^n \to \mathbb{R}^m$ is called *differentiable at* $x = \vec{a}$ if there exists a linear transformation $T_{\vec{a}} : \mathbb{R}^n \to \mathbb{R}^m$ such that

$$\lim_{\vec{h}\to 0} \frac{\left\| f(\vec{a}+\vec{h}) - f(\vec{a}) - T_{\vec{a}}(\vec{h}) \right\|}{\|\vec{h}\|} = 0.$$

Then $T_{\vec{a}}$ is called the *total derivative* at $x = \vec{a}$, and is often denoted by $Df(\vec{a})$.

Again, this says that $Df(\vec{a})$ is the best linear approximation to the function $f - f(\vec{a})$ at $\vec{a}$, in the sense that the difference $f(\vec{a}+\vec{h}) - f(\vec{a}) - Df(\vec{a})(\vec{h})$ is small compared to $\|\vec{h}\|$ as $\vec{h} \to 0$.

Proposition 4.10. *Suppose there exist two linear transformations*

$$T_{\vec{a}}^{(1)},\ T_{\vec{a}}^{(2)} : \mathbb{R}^n \to \mathbb{R}^m$$

such that

$$\lim_{\|\vec{h}\|\to 0} \frac{\|f(\vec{a}+\vec{h}) - f(\vec{a}) - T_{\vec{a}}^{(j)}(\vec{h})\|}{\|\vec{h}\|} = 0, \qquad j = 1, 2.$$

Prove that

$$T_{\vec{a}}^{(1)} = T_{\vec{a}}^{(2)}.$$

Proof. Since both limits are zero,

$$T_{\vec{a}}^{(2)}(\vec{h}) - T_{\vec{a}}^{(1)}(\vec{h}) = \{f(\vec{a}+\vec{h}) - f(\vec{a}) - T_{\vec{a}}^{(1)}(\vec{h})\} - \{f(\vec{a}+\vec{h}) - f(\vec{a}) - T_{\vec{a}}^{(2)}(\vec{h})\}.$$

Taking norms and applying the triangle inequality gives

$$\|(T_{\vec{a}}^{(2)} - T_{\vec{a}}^{(1)})(\vec{h})\| \leq \|f(\vec{a}+\vec{h}) - f(\vec{a}) - T_{\vec{a}}^{(1)}(\vec{h})\| + \|f(\vec{a}+\vec{h}) - f(\vec{a}) - T_{\vec{a}}^{(2)}(\vec{h})\|.$$

Dividing by $\|\vec{h}\|$ and taking the limit yields

$$\lim_{\|\vec{h}\|\to 0} \frac{\|(T_{\vec{a}}^{(2)} - T_{\vec{a}}^{(1)})(\vec{h})\|}{\|\vec{h}\|} = 0.$$

Let

$$L := T_{\vec{a}}^{(2)} - T_{\vec{a}}^{(1)}.$$

Then L is linear and

$$\lim_{\|\vec{h}\|\to 0} \frac{\|L(\vec{h})\|}{\|\vec{h}\|} = 0.$$

This only tells us something about vectors $\vec{h}$ near $\vec{0}$. It does not immediately tell you what $L(\vec{h})$ is for a fixed nonzero $\vec{h}$. Uniqueness needs a statement like the operator is zero on every vector. The trick plug in $\alpha\vec{h}$ where $\alpha \to 0$ is exactly how you turn a local limit statement into a global conclusion using linearity.
Let $\vec{h} \neq 0$ be fixed. For any scalar $\alpha \to 0$, we have $\alpha\vec{h} \to 0$. Thus the limit applies:

$$0 = \lim_{\alpha\to 0} \frac{\|L(\alpha\vec{h})\|}{\|\alpha\vec{h}\|}.$$

By linearity,

$$L(\alpha\vec{h}) = \alpha L(\vec{h}),$$

so

$$\frac{\|L(\alpha\vec{h})\|}{\|\alpha\vec{h}\|} = \frac{\|\alpha L(\vec{h})\|}{|\alpha|\,\|\vec{h}\|} = \frac{\|L(\vec{h})\|}{\|\vec{h}\|}.$$

$$0 = \lim_{\alpha\to 0} \frac{\|L(\alpha\vec{h})\|}{\|\alpha\vec{h}\|} = \lim_{\alpha\to 0} \frac{\|L(\vec{h})\|}{\|\vec{h}\|}.$$

Notice the quantity inside the limit is constant in α. The limit of a constant sequence/function is just the constant itself. So

$$\frac{\|L(\vec{h})\|}{\|\vec{h}\|} = 0 \implies \|L(\vec{h})\| = 0.$$

And the only vector with zero norm is the zero vector. Hence $L(\vec{h}) = 0$. Since the above holds for every nonzero vector $\vec{h}$, $L \equiv 0$. Thus

$$T_{\vec{a}}^{(2)} = T_{\vec{a}}^{(1)}.$$

□

Example 4.11. Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ defined by

$$f(x, y) = x^2 - y^2$$

Find the derivative of f at the point $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$, find the value of $Df\begin{pmatrix} 2 \\ 3 \end{pmatrix}$.

Solution. Let

$$Df\begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} a & b \end{pmatrix}$$

Using the definition of differentiability, let $h = (h_1, h_2) \to (0,0)$. Then

$$\lim_{\|\vec{h}\|\to 0} \frac{\left\| f\left(\begin{pmatrix} 2 \\ 3 \end{pmatrix} + \begin{pmatrix} h_1 \\ h_2 \end{pmatrix}\right) - f\begin{pmatrix} 2 \\ 3 \end{pmatrix} - \begin{pmatrix} a & b \end{pmatrix}\begin{pmatrix} h_1 \\ h_2 \end{pmatrix} \right\|}{\|\vec{h}\|}$$
$$= \lim_{\|\vec{h}\|\to 0} \frac{\left\| f\begin{pmatrix} 2+h_1 \\ 3+h_2 \end{pmatrix} - f\begin{pmatrix} 2 \\ 3 \end{pmatrix} - ah_1 - bh_2 \right\|}{\sqrt{h_1^2 + h_2^2}}.$$

Compute:

$$f(2+h_1, 3+h_2) - f(2,3) = (2+h_1)^2 - (3+h_2)^2 - \left(2^2 - 3^2\right).$$

and

$$(2+h_1)^2 - (3+h_2)^2 - 2^2 + 3^2 = (4 + 4h_1 + h_1^2) - (9 + 6h_2 + h_2^2) - 4 + 9 = 4h_1 - 6h_2 + h_1^2 - h_2^2.$$

So the numerator becomes

$$\left|(4-a)h_1 - (6+b)h_2 + h_1^2 - h_2^2\right|.$$

To make the linear terms vanish (why?), choose

$$4 - a = 0 \Rightarrow a = 4, \qquad -(6+b) = 0 \Rightarrow b = -6.$$

Then the limit reduces to

$$\lim_{(h_1,h_2)\to(0,0)} \frac{|h_1^2 - h_2^2|}{\sqrt{h_1^2 + h_2^2}} \le \lim_{(h_1,h_2)\to(0,0)} \frac{h_1^2 + h_2^2}{\sqrt{h_1^2 + h_2^2}} = \lim_{(h_1,h_2)\to(0,0)} \sqrt{h_1^2 + h_2^2} = 0.$$

Hence f is differentiable at $(2,3)$ and

$$Df\begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 & -6 \end{pmatrix}.$$

Example 4.12. For $f : \mathbb{R}^2 \to \mathbb{R}^2$ given by

$$f(x,y) = \begin{pmatrix} xy \\ y^2 \end{pmatrix},$$

find the value of $Df\begin{pmatrix} a \\ b \end{pmatrix}$.

Solution: For $f : \mathbb{R}^2 \to \mathbb{R}^2$ given by

$$f(x, y) = \begin{pmatrix} xy \\ y^2 \end{pmatrix},$$

the derivative at a point $\vec{a} = \begin{pmatrix} a \\ b \end{pmatrix}$ is the linear map $Df \begin{pmatrix} a \\ b \end{pmatrix}$ satisfying

$$f \begin{pmatrix} a+h \\ b+k \end{pmatrix} - f \begin{pmatrix} a \\ b \end{pmatrix} = Df_{\vec{a}} \begin{pmatrix} h \\ k \end{pmatrix} + o(\|(h, k)\|).$$

We will see this definition can be used to compute the derivative without using the limit definition directly. Compute the increment:

$$f \begin{pmatrix} a+h \\ b+k \end{pmatrix} = \begin{pmatrix} (a+h)(b+k) \\ (b+k)^2 \end{pmatrix} = \begin{pmatrix} ab + ak + bh + hk \\ b^2 + 2bk + k^2 \end{pmatrix}.$$

So

$$f \begin{pmatrix} a+h \\ b+k \end{pmatrix} - f \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} ak + bh + hk \\ 2bk + k^2 \end{pmatrix}.$$

The linear part in (h, k) is

$$\begin{pmatrix} bh + ak \\ 2bk \end{pmatrix},$$

and the remainder is

$$\begin{pmatrix} hk \\ k^2 \end{pmatrix},$$

(you can skip to verify the error term is $o(\|(h, k)\|)$, but we will verify it for the sake of completeness) whose norm is

which is $o(\|(h, k)\|)$ because $|hk| \leq \frac{1}{2}(h^2 + k^2)$ and $|k^2| \leq h^2 + k^2$.

Claim:	$\sqrt{x^2 + y^2} \leq \lvert x\rvert + \lvert y\rvert$.
Proof:	$(\lvert x\rvert + \lvert y\rvert)^2 = x^2 + y^2 + 2\lvert x\rvert\lvert y\rvert \geq x^2 + y^2$.
	Since both sides are nonnegative, taking square roots gives
	$\sqrt{x^2 + y^2} \leq \lvert x\rvert + \lvert y\rvert$.

$$\|R(h, k)\| = \sqrt{(hk)^2 + (k^2)^2} \leq |hk| + |k^2| \leq \frac{h^2 + k^2}{2} + (h^2 + k^2) = \frac{3}{2}(h^2 + k^2) = \frac{3}{2}\|(h, k)\|^2.$$

$$\frac{\|R(h, k)\|}{\|(h, k)\|} \leq \frac{3}{2}\|(h, k)\| \to 0,$$

Therefore,

$$Df_{\vec{a}} \begin{pmatrix} h \\ k \end{pmatrix} = \begin{pmatrix} bh + ak \\ 2bk \end{pmatrix},$$

So the matrix of $Df_{\vec{a}}$ is

$$Df_{\vec{a}} = \begin{pmatrix} b & a \\ 0 & 2b \end{pmatrix}.$$

Proposition 4.13. *If the function $f : U \subseteq \mathbb{R}^n \to \mathbb{R}$ is differentiable at $\vec{a}$, then the partial derivatives $\frac{\partial f}{\partial x_i}(\vec{a})$ exists for all $i = 1, 2, \ldots, n$ and $Df(\vec{a})$ is given by the row vector*

$$Df(\vec{a}) = \begin{pmatrix} \frac{\partial f}{\partial x_1}(\vec{a}) & \frac{\partial f}{\partial x_2}(\vec{a}) & \cdots & \frac{\partial f}{\partial x_n}(\vec{a}) \end{pmatrix}.$$

Proof. f is differentiable at $\vec{a}$ means that there exists a linear transformation $L \equiv Df(\vec{a}) : \mathbb{R}^n \to \mathbb{R}$ such that

$$f(\vec{a} + \vec{h}) = f(\vec{a}) + L(\vec{h}) + o(\|\vec{h}\|).$$

where $\lim_{\|h\| \to 0} \frac{o(\|\vec{h}\|)}{\|\vec{h}\|} = 0$. Suppose,

$$L(\vec{h}) = L \begin{pmatrix} h_1 \\ h_2 \\ \vdots \\ h_n \end{pmatrix} = \begin{pmatrix} c_1 & c_2 & \cdots & c_n \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \\ \vdots \\ h_n \end{pmatrix} = c_1 h_1 + c_2 h_2 + \cdots + c_n h_n.$$

Now, we want to impose the definition in each of the coordinate directions. For $i = 1, 2, \ldots, n$, let $\vec{h} = t\vec{e}_i$ where $\vec{e}_i$ is the i-th standard basis vector. Then

$$\begin{aligned} f(\vec{a} + t\vec{e}_i) &= f(\vec{a}) + L(t\vec{e}_i) + o(\|t\vec{e}_i\|) \\ f(\vec{a} + t\vec{e}_i) - f(\vec{a}) &= tL(\vec{e}_i) + o(|t|) \\ \frac{f(\vec{a} + t\vec{e}_i) - f(\vec{a})}{t} &= L(\vec{e}_i) + \frac{o(|t|)}{t} \\ \lim_{t \to 0} \frac{f(\vec{a} + t\vec{e}_i) - f(\vec{a})}{t} &= L(\vec{e}_i) + \lim_{t \to 0} \frac{o(|t|)}{t} \\ \frac{\partial f}{\partial x_i}(\vec{a}) &= L(\vec{e}_i) = c_i. \end{aligned}$$

□

Proposition 4.14. *If the function $f : U \subseteq \mathbb{R}^n \to \mathbb{R}^m$ is differentiable at $\vec{a}$, then the partial derivatives $\frac{\partial f^j}{\partial x_i}(\vec{a})$ exists for all $i = 1, 2, \ldots, n$, $j = 1, 2, \ldots, m$ and $Df(\vec{a})$ is given by the matrix*

$$Df(\vec{a}) = \begin{pmatrix} \frac{\partial f^1}{\partial x_1}(\vec{a}) & \frac{\partial f^1}{\partial x_2}(\vec{a}) & \cdots & \frac{\partial f^1}{\partial x_n}(\vec{a}) \\ \frac{\partial f^2}{\partial x_1}(\vec{a}) & \frac{\partial f^2}{\partial x_2}(\vec{a}) & \cdots & \frac{\partial f^2}{\partial x_n}(\vec{a}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f^m}{\partial x_1}(\vec{a}) & \frac{\partial f^m}{\partial x_2}(\vec{a}) & \cdots & \frac{\partial f^m}{\partial x_n}(\vec{a}) \end{pmatrix}.$$

Proposition 4.15. *Let $f : \mathbb{R}^k \to \mathbb{R}^n$ be a differentiable map. For each $i \in \{1, \ldots, n\}$ let*

$$\pi^i : \mathbb{R}^n \to \mathbb{R}, \qquad \pi^i(a^1, \ldots, a^n) = a^i$$

be the i-th coordinate projection, and define the i-th component function of f by

$$f^i := \pi^i \circ f : \mathbb{R}^k \to \mathbb{R}.$$

Then for every $a \in \mathbb{R}^k$,

$$D(f^i)(a) = \pi^i \circ Df(a).$$

Equivalently, as linear maps $\mathbb{R}^k \to \mathbb{R}$,

$$D(f^i)(a)[h] = \pi^i\big(Df(a)[h]\big) \qquad \text{for all } h \in \mathbb{R}^k.$$

Proof. See the proof from `https://math.stackexchange.com/a/4214572/1167703`. □

Proposition 4.16. *If $f : A \subseteq \mathbb{R}^n \to \mathbb{R}^m$ is differentiable at x_0, then f is continuous at x_0.*

Proof. By definition of the derivative of f we know that the limit

$$\lim_{h\to 0} \frac{||f(x_0+h) - f(x_0) + Dh||}{||h||} = 0,$$

for some linear transformation D. Therefore since this limit exists, the following is true:

$$0 = \left(\lim_{h\to 0} ||h||\right)\left(\lim_{h\to 0} \frac{||f(x_0+h) - f(x_0) + Dh||}{||h||}\right) = \lim_{h\to 0} ||f(x_0+h) - f(x_0) + Dh||.$$

By the triangle inequality,

$$\lim_{h\to 0} ||f(x_0+h) - f(x_0)|| \le \lim_{h\to 0} ||f(x_0+h) - f(x_0) + Dh|| + \lim_{h\to 0} ||Dh|| = \lim_{h\to 0} ||Dh||.$$

But since D is linear, it is continuous and $D0 = 0$, so the lattermost limit is just 0. Thus 0 is an upper bound for the limit, but since the norm is always non-negative, this implies that the limit is in fact 0. □

Example 4.17.

$$f(x,y,z) = x^2 + y^2 + z^2.$$

Take

$$\vec{a} = (1,1,1).$$

$$Df(x,y,z) = (2x, 2y, 2z).$$

$$Df(1,1,1) = (2,2,2).$$

Thus, for $\vec{v} = (v_1, v_2, v_3)$,

$$Df(1,1,1)(\vec{v}) = 2v_1 + 2v_2 + 2v_3.$$

$$D_{\vec{v}}f(1,1,1) = \lim_{t\to 0} \frac{(1+tv_1)^2 + (1+tv_2)^2 + (1+tv_3)^2 - 3}{t}.$$

Expanding,

$$(1+tv_1)^2 + (1+tv_2)^2 + (1+tv_3)^2 = 3 + 2t(v_1+v_2+v_3) + O(t^2).$$

Therefore,

$$D_{\vec{v}}f(1,1,1) = \lim_{t\to 0} \frac{3 + 2t(v_1 + v_2 + v_3) + O(t^2) - 3}{t} = 2(v_1 + v_2 + v_3).$$

Hence,

$$D_{\vec{v}}f(\vec{a}) = Df(\vec{a})(\vec{v}).$$

If f is differentiable at $\vec{a}$, then

$$\begin{aligned} f(\vec{a} + \vec{h}) &= f(\vec{a}) + Df(\vec{a})(\vec{h}) + o(|\vec{h}|) \\ \overset{\vec{h}=t\vec{v}}{\Longrightarrow} \frac{f(\vec{a} + t\vec{v}) - f(\vec{a})}{t} &= Df(\vec{a})(\vec{v}) + \frac{o(t|\vec{v}|)}{t}. \end{aligned}$$

As $t \to 0$, the error term goes to 0, giving

$$D_{\vec{v}}f(\vec{a}) = Df(\vec{a})(\vec{v}).$$

Proposition 4.18. *When f is differentiable at $\vec{a}$, for any $\vec{v}$, the directional derivative of f at $\vec{a}$ in the direction $\vec{v}$ exists and is given by*

$$D_{\vec{v}}f(\vec{a}) = Df(\vec{a})(\vec{v}).$$

Proof. See the proof from Theodore Shifrin (Chapter 3, section 2, proposition 2.3). □

Proposition 4.19. *If $f : U \subseteq \mathbb{R}^n \to \mathbb{R}^m$ has continuous partial derivatives, then f is differentiable.*

Proof. See the proof from Theodore Shifrin (Chapter 3, section 2, proposition 2.4). □